BTP Project

Multi-Strategy AVOA
→ Multi-Objective MIAVOA

Plan of Work · Phase-wise Execution · Novelty & Evaluation

B.Tech Project | Team Presentation

Project Roadmap

Three-phase approach from baseline to novelty

PHASE 1

Reproduce MIAVOA

Implement all 4 improvement strategies from the base paper into AVOA — one per teammate

PHASE 2

Novelty MO-MIAVOA

Extend MIAVOA to multi-objective using Pareto dominance, archive, crowding distance

PHASE 3

Benchmarking & Comparison

Compare MO-MIAVOA against NSGA-II, MOEA/D, MOPSO on IGD, HV, GD, Spread

AVOA → MIAVOA (reproduce) → MO-MIAVOA (novelty) → Compare

PHASE 1

Implementing the 4 MIAVOA Strategies

01 Gaussian Quasi-Reflection Learning (GQRL)

Initialization

Combines Gaussian chaos mapping with quasi-reflection-based learning to boost initial population diversity and randomness. Prevents premature convergence early on.

02 Adaptive Control Strategy (ACS)

Exploitation

Dynamically adjusts influence weights (α_1, α_2) of best and second-best vultures across iterations — more global search early, more local exploitation later.

03 Elite Candidate Pool Strategy (ECPS)

Exploration

In spiral-flight phase, generates 4 candidate positions (R1–R4) from S1/S2 and picks the best. Widens discovery field and helps escape local optima.

04 Improved Hunger Parameter (IHP)

Balance

Modifies the starvation factor F with an exponential decay term. Ensures smooth, well-calibrated balance between exploration and exploitation throughout iterations.

TEAM TASK DIVISION

Each teammate owns one MIAVOA strategy — implement & validate independently

T1

Strategy 1: GQRL

Gaussian chaos mapping + quasi-reflection initialization. Code AVOA1, test on F1–F7.

T2

Strategy 2: ACS

Adaptive weight control for α_1/α_2 across iterations. Code AVOA2, test on CEC2022.

T3

Strategy 3: ECPS

Elite candidate pool in spiral-flight phase. Code AVOA3, verify local optima escape.

T4

Strategy 4: IHP

Exponential hunger parameter modification. Code AVOA4, measure exploration/exploitation balance.

✔ After individual validation, all 4 strategies are merged to reproduce MIAVOA and cross-checked against the paper's results.

PHASE 2

Our Novelty: MO-MIAVOA

Multi-Objective MIAVOA

Idea from paper's own future work suggestion: "MIAVOA will be extended and applied to multi-objective optimization."

1

Pareto Dominance

Replace single fitness comparison with Pareto dominance check. A solution dominates another if it is no worse in all objectives and better in at least one.

2

External Archive

Maintain a bounded archive of non-dominated solutions (the Pareto front approximation) throughout iterations. Archive is updated each generation.

3

Crowding Distance

Use crowding distance to preserve diversity on the Pareto front. Prefer solutions in less-crowded regions when archive is full.

4

Leader Selection

Vulture leaders (BestVulture1/2) are selected from the archive using tournament or roulette based on crowding distance instead of scalar fitness.

PHASE 3

Benchmarking & Comparison of MO-MIAVOA

Comparison Algorithms

NSGA-II	Non-dominated Sorting GA — gold standard for MO
MOEA/D	Decomposition-based; strong on many-objective
MOPSO	Multi-objective Particle Swarm Optimization
MOGWO	Multi-objective Grey Wolf Optimizer (peer MO)
MOAVOA	Baseline MO extension of plain AVOA

Performance Metrics

IGD	Inverted Generational Distance — proximity + diversity of found front
HV	Hypervolume — volume dominated by the Pareto front approximation
GD	Generational Distance — convergence to true Pareto front
Spread	Distribution uniformity across the Pareto front solutions
Runtime	Computational time — confirms complexity stays $O(N^2T)$

Test suites: ZDT1–ZDT6 | DTLZ1–DTLZ4 | Engineering multi-objective problems

Summary

What we will deliver

Phase 1

- Reproduce all 4 MIAVOA strategies in code
- Each teammate validates their strategy on benchmark functions
- Merge into full MIAVOA and match paper results

Phase 2

- Integrate Pareto dominance into MIAVOA fitness logic
- Add external archive + crowding distance management
- Modify leader selection for multi-objective context

Phase 3

- Run MO-MIAVOA on ZDT, DTLZ test suites
- Compare with NSGA-II, MOEA/D, MOPSO, MOGWO, MOAVOA
- Evaluate on IGD, HV, GD, Spread — report results